Physical Conditioning and Quantitative Interfaces for Bi-Lipschitz Coordinates on Noncollapsed RCD Spaces

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Abstract

The source manuscript constructs open bi-Lipschitz coordinates near each prescribed regular point of a noncollapsed RCD space by summable directional pinching and renormalization in an unbounded, scale-dependent physical metric. We study two quantitative interfaces without replacing that global construction. First, a semigroup representation of its two-center commutator replaces a small operator-norm hypothesis by a spectral-width condition on the symmetric part, with a sharp inverse bound and explicit coefficient recovery. Second, simultaneous optimization of the Taylor remainder and the two coordinate tails gives a sharp minimax bound for the stated dyadic scalar envelope, including domain clearance and a finite resolution cutoff. We also derive physical first- and second-derivative budgets for an inexact inverse-transpose update, without claiming that these finite budgets close the source's all-order induction. Exact noncommuting fixtures and counterexamples distinguish finite conditioning, all-scale control, and an actual open chart. Contribution assessment: C2/5, the proved interface refinements are modest quantitative extensions and do not replace the source's geometric pinching or infinite-scale chart theorem.

1 Problem, normalization, and source boundary

Let $(X, d, \mathcal{H}^n)$ be noncollapsed RCD(K, n), with integer $n \geq 2$. Here the measure is exactly n-dimensional Hausdorff measure, the metric is complete, separable and geodesic with full support, and the RCD convention is the full curvature-dimension condition together with quadratic Cheeger energy. A prescribed point p is n-regular when every pointed metric tangent there is Euclidean $\mathbb{R}^n$. The source [1] claims a dimension-dependent L_n and an open neighborhood U of p admitting an open-image homeomorphism $F : U \to \mathbb{R}^n$ such that

$$L_n^{-1} d(x, z) \leq |F(x) - F(z)| \leq L_n d(x, z) \quad \text{for every } x, z \in U.$$

The distance is the restricted *ambient* distance, not an intrinsic path distance on U. The radius depends on the space and the point. This is stronger than almost-everywhere rectifiability by measurable pieces. It does not assert a smooth metric, a bounded Laplacian of F, or that the exact regular locus is open.

All source text has been read for this study, but reading a proof is not an independent verification of all its imported RCD analysis. The source uses measured convergence, harmonic compactness, measure-valued Bochner calculus, and a simultaneous all-order induction. Those inputs are not

reproved here. Our local statements are independent finite-dimensional or elementary analytic deductions with their assumptions displayed. No finite test below establishes the claimed RCD theorem.

1.1 The physical metric is not optional

At dyadic radius $r_i = 2^{-i}$ the source has a harmonic column u and

$$G = (\langle \nabla u_j, \nabla u_k \rangle), \qquad R_i(y) = \frac{1}{\mathcal{H}^n(B_{r_i}(y))} \int_{B_{r_i}(y)} G \, d\mathcal{H}^n > 0.$$

Its positive matrix-valued Bochner measure is $M = \frac{1}{2}\Delta G + \kappa G \mathcal{H}^n$ after rescaling the curvature lower bound to $-\kappa$. With an interior cutoff χ, corrected heat averages and their inverses are

$$Q_i = P_{r_i^2}(\chi G) + \lambda r_i^2 I, \qquad H_i = Q_i^{-1}. \tag{1}$$

The correction $\lambda \downarrow 0$ is chosen to dominate curvature and cutoff errors. The source proves relative comparison $Q_i = (1 + o(1))R_i$, monotonicity $H_i \preceq H_{i+1}$, and slow power growth, not boundedness of H_i as $i \to \infty$. Vectors have norm $|\xi|_{H_i}$ and covectors have norm $|\eta|_{Q_i}$. Interchanging these norms or imposing a Euclidean spectral floor loses the source problem.

For a fixed enlargement L the directional matrices are

$$D_{i,y} = R_i(y)^{-1} \big[r_i^{2-n} M(B_{Lr_i}(y)) \big] R_i(y)^{-1}.$$

Their central estimate is

$$D_{i,y}[u(x) - u(z)] \le C r_i^2 \big(E_i(x) + E_i(z) \big), \qquad \sup_x \sum_{i \ge 0} E_i(x) \longrightarrow 0. \tag{2}$$

It holds simultaneously at every scale and for the specified nearby centers, with enlargements fixed before the late rescaling. Summability of $\sqrt{E_i}$ is neither assumed nor generally true.

1.2 The proof route and its sensitive transitions

The source route, with literal module line locators, is as follows.

1. `coordinates.tex:1--465`: local weak calculus and noncollapsed trace identity; uniformly flat rescalings; harmonic normalization; open harmonic coordinates and interior value-space ellipsoids.

2. `coordinates.tex:467--1166`: continuous heat representatives against possibly singular measures, monotone Q_i, noncommuting inverse increments, change of normalization, cumulative directional matrix masses.

3. `pinching.tex:97--353`: positive-source Dirichlet estimates and squared-distance Poisson replacements. The mass is controlled by telescoping volume drops, not by a sum of square-root errors.

4. `pinching.tex:364--654`: weak radial cancellation, harmonic affine improvement, exponentially smoothed error budgets and positive floors for choosing fixed patch centers.

5. `pinching.tex:701--1149`: two-center commutator, radial transport *before* normalization of a small residual, sharp affine projection by compactness, and the full directional Bochner mass.

6. `renormalization.tex:112--685`: Gevrey patch fields, the inverse-transpose iteration, all-order inverse bounds without a new exponential base, and diagonal second-derivative square summability.

7. `renormalization.tex:697--839`: fixed-scale uniform tails, pair-adapted Taylor segments, all-pairs ambient-distance comparison and invariance of domain.

Our first refinement applies to item 5, the second to item 7, and the inexact-update account explains an additional requirement in item 6. They do not bypass items 1–4 or the source's compactness arguments.

2 A rotation-insensitive two-center bound

All matrices in this section are real n by n and $\|\cdot\|$ denotes Euclidean operator norm. Set

$$C_s = \operatorname{sym} C = \tfrac{1}{2}(C + C^T), \qquad \omega(C) = \lambda_{\max}(C_s) - \lambda_{\min}(C_s).$$

Thus $\omega(C) \le 2\|C\|$, but it ignores a scalar multiple of I and a purely skew-symmetric part.

Theorem 2.1 (Width-controlled commutator inversion)**.** *If $\omega(C) < 1$, the real-linear map $\mathcal{T}_C(B) = B + BC - CB$ is invertible and*

$$\mathcal{T}_C^{-1}(F) = \int_0^\infty e^{-t} e^{tC} F e^{-tC}\, dt, \qquad \|\mathcal{T}_C^{-1}(F)\| \le \frac{\|F\|}{1 - \omega(C)}. \tag{3}$$

The factor $(1-\omega)^{-1}$ is best possible using only the number $0 \le \omega < 1$, already for real diagonal two by two matrices.

Proof. Differentiating $|e^{tC}v|^2$ gives $\|e^{tC}\| \le e^{t\lambda_{\max}(C_s)}$. Applying the same argument to $-C$ gives $\|e^{-tC}\| \le e^{-t\lambda_{\min}(C_s)}$. Consequently the integrand has norm at most $e^{-(1-\omega)t}\|F\|$, so the integral converges absolutely. If $Y(t) = e^{-t}e^{tC}Fe^{-tC}$, then $Y' = -Y + CY - YC = -\mathcal{T}_C Y$. Integrating yields $\mathcal{T}_C \int_0^\infty Y = F$. A right inverse for a linear map between equal finite dimensions is an inverse. Integration of the norm bound proves the estimate.

For sharpness take $C = \operatorname{diag}(\omega/2, -\omega/2)$ and $B = e_1 e_2^T$. Then $\mathcal{T}_C B = (1-\omega)B$. At $\omega = 1$ this nonzero matrix lies in the kernel. □

This is a classical semigroup proof of a Sylvester inverse estimate, specialized to the exact operator used by the source. It is not a new general matrix inversion principle.

Corollary 2.2 (Certified uncertainty in the radial coefficient)**.** *Suppose $\|\operatorname{sym}(C - \widehat{C})\| \le \eta$ and $\widehat{\omega} = \omega(\widehat{C})$ satisfies $\widehat{\omega} + 2\eta < 1$. If $\mathcal{T}_C B = F$ and $\|F - \widehat{F}\| \le \delta$, then*

$$\|B\| \le \frac{\|\widehat{F}\| + \delta}{1 - \widehat{\omega} - 2\eta}.$$

No estimate on the skew part of $C - \widehat{C}$ is needed for this bound, but the supplied residual F must be the residual of the actual C.

Proof. The maximum symmetric eigenvalue increases by at most η and the minimum decreases by at most η. Apply Theorem 2.1. Computing a residual for $\widehat{C}$ alone is not the stated premise. □

2.1 Application to the actual two-center algebra

In the source, equal-source Poisson replacements yield radial coefficient matrices A_x, A_z. Put $C = A_z - I$, $B = A_z - A_x$, $\delta = U(x) - U(z)$ and $c = A_x\delta$. The weak commutator and affine coefficient extraction provide

$$F = BA_z - (A_z - I)B = B + BC - CB, \qquad Cc,$$

with bounds of order $s = \sqrt{E_i(x) + E_i(z)}$. The source absorbs $2\|C\|\,\|B\|$ when $\|C\| < 1/4$. The following replaces only this algebraic absorption and keeps the subsequent recovery of the actual coordinate difference.

Proposition 2.3 (Coefficient recovery with named constants)**.** *Let a finite-measure domain V have measure $m_V > 0$. Suppose*

$$h = BU + c + e, \quad \|e\|_{L^2(V)} \le e_0, \quad \|U\|_{L^2(V;\mathbb{R}^n)} \le U_0,$$

$c = A_x\delta$, $\|A_x^{-1}\| \le M_0$, $\|F\| \le f_0$, $\|Cc\| \le g_0$, and $\omega(C) < 1$. Writing $b_0 = f_0/(1 - \omega(C))$, one has

$$\|h - \delta\|_{L^2(V)} \le b_0U_0 + \sqrt{m_V}M_0(g_0 + b_0|c|) + e_0. \tag{4}$$

Proof. Theorem 2.1 gives $\|B\| \le b_0$. Because $A_x - I = C - B$,

$$\delta - c = -A_x^{-1}(A_x - I)c = -A_x^{-1}(C - B)c.$$

Hence $|\delta - c| \le M_0(g_0 + b_0|c|)$. The triangle inequality in $L^2(V)$ proves (4). □

If $f_0, g_0, e_0 = O(s)$ and the other constants remain bounded, (4) is $O(s)$, exactly the scale needed before squaring to obtain the directional Bochner estimate. The hypothesis $\|A_x^{-1}\| \le M_0$ is not removed. Nor does this proposition construct radial identities when A_x, A_z are far from I: their construction and the other geometric uses of closeness remain source inputs.

2.2 Sharpness and negative controls

For $C = L\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\omega(C) = 0$ for every L, and the inverse bound is one. The source's small-norm absorption cannot accept large L, while this local inverse can. Equality occurs on $F = I$. Thus rotations are not themselves a commutator conditioning loss.

Eigenvalue separation without nonnormality control is different. For $C = Le_1e_2^T$ both eigenvalues vanish, yet for $F = e_2e_1^T$,

$$\mathcal{T}_C^{-1}F = e_2e_1^T + L\,\mathrm{diag}(1, -1) - 2L^2e_1e_2^T.$$

The norm is at least $2L^2$ by its upper-right entry. Here $\omega(C) = |L|$ correctly prevents a uniform bound as L grows. The formula follows from $\mathcal{T}_C = I - \mathrm{ad}_C$ and $(\mathrm{ad}_C)^3 = 0$. It illustrates why mere invertibility, or a favorable list of eigenvalues, is not a physical error estimate.

3 Joint scale selection for all-pairs comparison

This section assumes an existing source construction. It does not infer a tail from finitely many maps. Let u be an open homeomorphism onto an open value domain $\Omega \subset \mathbb{R}^n$, let $h_i \in C^2(\Omega; \mathbb{R}^n)$, and let $H_i(x) > 0$. Assume $F = \lim h_i \circ u$ exists and

$$|F(x) - h_i(u(x))| \le \tau r_i, \qquad r_i = 2^{-i}, \quad \tau \ge 0. \tag{5}$$

For a pair x, z, write $t = d(x, z) > 0$, $\Delta = u(z) - u(x)$ and $s = r_i/t$. The following are assumptions at the *chosen scale*:

$$at \le |\Delta|_{H_i(x)} \le bt, \quad a, b > 0, \tag{6}$$

$$m|\xi|_{H_i(x)} \le |Dh_i(u(x))\xi| \le M|\xi|_{H_i(x)}, \quad m > 0, \tag{7}$$

$$|h_i(u(z)) - h_i(u(x)) - Dh_i(u(x))\Delta| \le \frac{K}{r_i}|\Delta|^2_{H_i(x)}, \quad K \ge 0. \tag{8}$$

For (8), the complete straight segment must lie in Ω with the physical second-derivative bound along it. A convenient sufficient domain gate is that the ellipsoid $\{P : |P - u(x)|_{H_i(x)} < cr_i\}$ lies in the admissible value domain and $s > b/c$. The source supplies this using preimages in an interior metric ball, not convexity of the whole image.

Theorem 3.1 (Coupled dyadic error envelope)**.** *Put $A = Kb^2$. For any $D > 0$ and a dyadic scale with $s \in [D, 2D]$ at which the preceding assumptions hold,*

$$(ma - \mathcal{E}(D))t \le |F(z) - F(x)| \le (Mb + \mathcal{E}(D))t, \tag{9}$$

where

$$\mathcal{E}(D) = \max\{A/D + 2\tau D,\ A/(2D) + 4\tau D\}. \tag{10}$$

If $A, \tau > 0$, the minimum over $D > 0$ is

$$D_* = \sqrt{\frac{A}{4\tau}}, \qquad \min \mathcal{E} = 3\sqrt{A\tau}. \tag{11}$$

With an additional lower restriction $D \ge D_0 > 0$, the minimum occurs at $\max(D_0, D_*)$. Strict domain restrictions require a strict increase if the minimizer equals the forbidden endpoint.

Proof. Taylor error divided by t is at most A/s. The two tails total at most $2\tau st$. Use the triangle inequality with the principal Jacobian term, whose length lies between mat and Mbt. The function $A/s + 2\tau s$ is convex on $s > 0$, hence its maximum on $[D, 2D]$ occurs at an endpoint, proving (10).

The difference of its two endpoint values is $A/(2D) - 2\tau D$. For $D \le D_*$ the first endpoint is dominant and strictly decreasing, since its derivative is $-A/D^2 + 2\tau < 0$ there. For $D \ge D_*$ the second is dominant and strictly increasing, since $-A/(2D^2) + 4\tau > 0$ there. At the crossing both equal $3\sqrt{A\tau}$. This proves the unconstrained and constrained assertions. □

Optimality here means the minimax value of the stated scalar envelope; it is not an optimal geometric distortion theorem. If $\tau = 0$, the envelope decreases to zero as $D \to \infty$, provided all other hypotheses can still be supplied. If $A = 0$, the best permitted scale is the smallest one. Changing D can change the geometric flatness threshold required for (6); that cost cannot be optimized away by treating a, b as unconditional constants.

Corollary 3.2 (Source constants and resolution)**.** *For the source values $a = 1/2$, $b = 2$, $m = A_0^{-1/2}$, $M = A_0^{1/2}$ and $K = K_T$, the unconstrained envelope is $6\sqrt{K_T\tau}$ and $D_* = \sqrt{K_T/\tau}$. It gives a positive lower Lipschitz bound when $6\sqrt{K_T\tau} < 1/(2\sqrt{A_0})$, subject to the domain and pair-normalization gates. If only levels $0, \dots, N$ and their actual tails (5) are certified, the dyadic selection covers the distance window*

$$\frac{2^{-N}}{2D} \le t \le \frac{1}{D},$$

with endpoint choices $s \in [D, 2D]$. It does not cover arbitrarily small distances at fixed N.

Proof. Substitute in Theorem 3.1. The intervals $[2^{-i}/(2D), 2^{-i}/D]$ for $0 \le i \le N$ meet at endpoints and their union is the displayed interval. □

For a fully explicit budget, take $a = 1/2, b = 2, m = 1/2, M = 2, K = 1, \tau = 1/2304$ and suppose the domain gate permits $D = 48$. Then $A = 4$, $D_* = 48$ and $\mathcal{E} = 1/8$. The resulting pair constants are $1/8$ and $33/8$. This is a conditional budget example, not a numerically constructed RCD chart.

4 What an inexact update must actually control

The source already proves the exact metric-step identity, the all-order inverse majorant, the dyadic half-factor in the diagonal recurrence and the uniform tail (5). They are not contributions of this report. For an implementation or a conditional finite approximation, however, the residual must be differentiated through the inverse transpose.

Proposition 4.1 (Frozen physical residual budgets)**.** *Let $h \in C^3(\Omega; \mathbb{R}^n)$, $J = Dh$ invertible, and $\eta \in C^2(\Omega; \mathbb{R}^n)$ a covector-valued correction residual. At the evaluation point fix $H > 0$, $Q = H^{-1}$ and a radius $r > 0$. Assume, in these frozen source units,*

$$\sigma_{\min}(JH^{-1/2}) \ge m > 0, \quad r\|D^2 h\|_{H,H\to\mathbb{R}^n} \le L_2, \quad r^2\|D^3 h\|_{H,H,H\to\mathbb{R}^n} \le L_3,$$

and

$$|\eta|_Q \le r a_0, \qquad \|D\eta\|_{H\to Q} \le a_1, \qquad r\|D^2\eta\|_{H,H\to Q} \le a_2, \qquad a_j \ge 0.$$

For $e = J^{-T}\eta$ the following pointwise estimates hold:

$$|e| \le r a_0/m, \tag{12}$$

$$\|De\|_{H\to\mathbb{R}^n} \le d_1 := a_1/m + L_2 a_0/m^2, \tag{13}$$

$$r\|D^2 e\|_{H,H\to\mathbb{R}^n} \le a_2/m + 2L_2 a_1/m^2 + (2L_2^2/m^3 + L_3/m^2)a_0. \tag{14}$$

Here matrix-valued H is frozen when derivatives are taken; there is no differentiation of the possibly nonsmooth field $x \mapsto H_i(x)$.

Proof. Make the constant affine change $P = P_0 + rH^{-1/2}X$ and set $\widetilde{h} = r^{-1}(h(P) - h(P_0))$, $\widetilde{\eta} = r^{-1}H^{-1/2}\eta(P)$. Its Jacobian is $\widetilde{J} = JH^{-1/2}$ and $\widetilde{J}^{-T}\widetilde{\eta} = r^{-1}J^{-T}\eta$. At zero, $\|\widetilde{J}^{-1}\| \le m^{-1}$, $\|D\widetilde{J}\| \le L_2$, and $\|D^2\widetilde{J}\| \le L_3$.

Write $K = \widetilde{J}^{-T}$. Differentiating the inverse identity gives

$$D_\xi K = -K(D_\xi \widetilde{J})^T K.$$

The mixed second derivative contains two terms with two first derivatives of $\widetilde{J}$ and one with its second derivative. Thus $\|DK\| \le L_2/m^2$ and $\|D^2K\| \le 2L_2^2/m^3 + L_3/m^2$. The product rule for $K\widetilde{\eta}$ has respectively one, two, and four terms when the two mixed first-derivative terms are counted separately. Substituting the bounds a_0, a_1, a_2 and undoing the constant coordinate change gives (12)–(14). □

Corollary 4.2 (One-step metric loss)**.** *Suppose the ideal finite update has derivative J_+ with $m_+|\xi|_H \le |J_+\xi| \le M_+|\xi|_H$. If the only change is addition of e as in Proposition 4.1, then*

$$(m_+ - d_1)|\xi|_H \le |(J_+ + De)\xi| \le (M_+ + d_1)|\xi|_H.$$

In particular invertibility is retained when $d_1 < m_+$, and

$$\left\| H^{-1/2}\big((J_+ + De)^T(J_+ + De) - J_+^T J_+\big)H^{-1/2} \right\| \le 2M_+ d_1 + d_1^2.$$

Proof. Use the triangle inequality and its lower form on each vector. The Gram difference consists of the two cross terms and $(De)^T De$; their normalized norms give the displayed bound. □

This is a one-step statement about the *actual* maps and physical derivatives. Along a perturbed iteration all these hypotheses must hold for the perturbed h_i, not merely for a separate exact sequence. In particular, (14) already needs D^3h. Closing an infinite all-order iteration requires residual bounds at all higher orders with the permitted factorial base, in addition to cumulative metric error and the common domain. We do not prove such a closure. If an actual sequence has uniform increment bounds $r_i b_i$, the value tail is at most $\sum_{j\geq i} r_j b_j$; that elementary summation does not supply derivative control or injectivity.

4.1 Two negative controls specific to the update

In dimension one take $h(P) = e^P$, $v(P) = 1$. At $P = 0$, $J = 1$, $Dv = 0$ but

$$D(J^{-T}v) = -1.$$

The update $h + J^{-T}v = e^P + e^{-P}$ has zero derivative there. Treating the inverse transpose as constant would falsely predict no derivative change. More generally the scalar product term in the source metric identity is exactly

$$|(J + DW)\xi|^2 - |J\xi|^2 = 2\xi^T Dv\, \xi - 2\langle D^2h(\xi,\xi), J^{-T}v\rangle + |DW\xi|^2.$$

The second term cannot be removed just because the value residual is small.

Conversely take $h(P) = P$ and $\eta(P) = a\sin(kP)$ in one coordinate, with identity in the remaining coordinates. Then $|\eta| \leq a$, but at zero the derivative error is ak. Arbitrarily small value error permits arbitrarily large derivative error. Neither a finite value grid nor a value-only tail can replace (13).

5 Worked physical examples and finite limits

5.1 Exact flat harmonic data

On Euclidean $\mathbb{R}^2$, take $u(x) = Lx$ for any invertible real L. Then $R_i = LL^T$ at every radius, the uncorrected heat Gram is the same, and $M(u) = 0$. With $H = (LL^T)^{-1}$,

$$|u(x) - u(z)|_H^2 = (x - z)^T L^T (LL^T)^{-1} L(x - z) = |x - z|^2.$$

For example $L = \left(\begin{smallmatrix} 1 & 1 \\ 0 & \epsilon \end{smallmatrix}\right)$ has a small singular value as $\epsilon \downarrow 0$, but the physical equality is exact. The linear whitening $h(P) = H^{1/2}P$ is an isometry up to an orthogonal factor. This is an actual smooth noncollapsed RCD example, not evidence for the nonsmooth construction.

5.2 Relative error survives anisotropy

For $G_\epsilon = \mathrm{diag}(1, \epsilon^2)$ and $\widehat{G}_\epsilon = \mathrm{diag}(1, \epsilon^2 + \epsilon)$, $\|\widehat{G}_\epsilon - G_\epsilon\| \to 0$, but the norm ratio in the second direction is $\sqrt{1 + \epsilon^{-1}}$. This elementary physical-metric obstruction is already present in the original Ram chapter [7]; it is not a new result.

Even increasing positive matrices can be unbounded: $H_i = (1 + i)^a I$ for any $a > 0$ have adjacent ratios tending to one and arbitrarily slow power growth relative to 2^i, but no finite limit. Appending this sequence after any chosen finite prefix preserves every test on that prefix. These matrices are not asserted to come from an RCD space. Their role is to refute an invalid inference from a finite prefix to a uniform spectral floor.

5.3 Geometric and measure-theoretic boundaries

A two-dimensional Euclidean cone of angle less than 2π is flat away from its vertex but has a non-Euclidean tangent at the vertex. Euclidean computations on finitely many small patches avoiding that vertex cannot certify its regularity. This is not a counterexample to existence of some bi-Lipschitz parametrization of that cone; it distinguishes regularity assumptions from local tests away from the point. Likewise, a singular Bochner measure cannot be integrated against an arbitrary almost-everywhere representative of a heat gradient. The source uses continuous kernel representatives and capacity-compatible weak tests before such pairings.

For $E_i = \epsilon/(i+1)^2$, $\sum_i E_i < \infty$ but $\sum_i \sqrt{E_i} = \infty$. A proof that replaces the source's square-sum convolution by an unweighted square-root sum fails on this simple sequence. Finally, nonsingular Jacobians of finite h_i are not by themselves global injectivity: the source proves all-pairs separation of the limit and then invokes invariance of domain on $u(B_\rho(p))$.

6 Ram proof comparison and permitted transfer

The selected original TR207 proof [6], lines 164–318, uses compactly supported C^N atoms on an interval, a declared bilinear pairing, and repeated integration by parts to transfer derivatives onto coefficient-weighted atoms. Its exact remainder contains every derivative of the taper and variable coefficient. That is a useful discipline for weak radial identities, but it does not authorize transplanting Euclidean partial derivatives to an RCD metric. Here the corresponding objects are the weak Laplacian, Sobolev test functions, and measure-valued Bochner calculus already established in the source.

TR207 lines 367–503 prove finite atom-bank identifiability and a perturbed-design bound with a singular-value margin. This is directly relevant to solving the finite coefficient equation $\mathcal{T}_C B = F$. Theorem 2.1 supplies a source-specific physical inverse estimate instead of merely counting equations. It does not reinterpret the infinitely many scales as a single finite atom bank.

The original physical-metric chapter [7], lines 1–210, proves that an injective synthesis J carries norm $c^* J^* J c$, and decomposes a full residual into a projected part and its orthogonal complement. It also shows why finitely many bounded measurements on an infinite-dimensional space cannot give a positive continuum lower frame bound. These proofs motivate the use of H_i, Q_i and the explicit unresolved scales in Corollary 3.2; they do not establish a geometric continuum completeness theorem.

The selected core-pat-16 body [8], lines 375–487, differentiates actual implicit and regularized solves, retaining matrix derivatives, and separates rank from inverse amplification. Proposition 4.1 makes that requirement explicit for the source's inverse-transpose update, including the second derivative. No probability covariance interpretation is imposed on the deterministic RCD Bochner measures.

7 Current public context and source status

The cached source is the September 25, 2026 OpenAI manuscript. The captured-tree comparison records unchanged bytes between the October 6 initial commit and the captured October 8 tree. A fresh October 9 query of the public GitHub path history returned that October 6 initial commit as its most recent path commit. This is provenance, not mathematical acceptance or an independent proof audit. No newer source was silently substituted.

The primary abstracts and version records consulted on October 9 give the following distinctions. Mondino–Naber [2] prove rectifiability and almost-everywhere Euclidean tangents; this does not by itself give an open chart at every prescribed regular point. Kapovitch–Mondino [3] establish topological regularity and stability for noncollapsed RCD spaces, up to a small singular set. Colding–Naber [4] discuss bi-Lipschitz embeddings of Reifenberg sets and examples excluding stronger differentiability. These are not grounds for adding a smooth metric conclusion to the source.

Honda–Zhang's August 17, 2026 version [5] assumes positive injectivity radius and obtains much stronger metric and differential regularity. That additional assumption cannot be dropped by citation. For these external papers our fresh access was to primary abstracts and version metadata, not their complete proofs. The report's local linear-algebra, Taylor and derivative arguments are proved here and do not rely on an uninspected external proof of them. The literature check is not an exhaustive priority search.

8 Benefits, costs, and further work

Interface	Retained or improved account	Unreplaced obligation
Two-center commutator	Width $\omega(C)$ instead of $2\|C\|$; explicit recovery cost	Actual radial identities, affine extraction, A_x^{-1} bound
Pair comparison	Coupled dyadic envelope and exact minimizer	Full segment clearance, scale-relative harmonic estimates and actual tails
Inexact correction	Value and first/second derivative residuals in physical units	All-order residual majorants and cumulative perturbed bootstrap
Finite moments	Declared metric and identified coefficient space	Continuum complement and every-scale, every-center quantifiers

The first refinement is advantageous when a radial coefficient has large rotational or scalar content but small symmetric width. Its limitation is the possible approach to width one and the independent cost of A_x^{-1}. The second uses the same scale for Taylor error and both tails rather than maximizing the two errors separately. It can sharpen explicit budgets, but a smaller error may require a larger scale ratio and a later geometric rescaling. Neither claim is a measured runtime improvement.

A natural further project is an inexact all-order theorem with a Gevrey residual bank whose derivative base is compatible with the source induction. Another is a computable geometric modulus for the uniform compactness steps, including the full Bochner measure and the extension of an unnormalized residual to the larger projection ball. Both require new hypotheses or analysis; the finite results here do not provide them. Width-controlled two-center estimates could also be studied in a different normalization, but its metric and the conditioning of the conjugation must be transported explicitly.

9 Evidence scope and conclusion

Fresh numbered, bounded displays cover all seven assigned public text files: `paper.tex` (57 lines), `introduction.tex` (258), `coordinates.tex` (1,166), `pinching.tex` (1,149), `renormalization.tex` (839), `references.bib` (408), and the README (17), totaling 3,894 lines. The source PDF is not counted as an additional proof read. Private proof scope is TR207 lines 164–318 and 367–503, the

physical-metric chapter lines 1–210, and core-pat-16 lines 375–487: 615 distinct lines. Complements, pre/post hashes, and the individual display intervals are retained in the read ledger. The figure referenced by the metric chapter and its other chapter dependencies were not read by inclusion.

The accompanying owned checker exercises the commutator equation, width sharpness, non-normal controls, physical flat harmonic data, the dyadic minimax algebra, and inverse-transpose derivative identities. Its result distinguishes exact cases from floating diagnostics and counts assertions separately. These are finite algebraic/manufactured tests, not RCD proof verification. The executed result is 93 exact cases with 241 assertions and, separately, 24 floating-point diagnostic cases with 48 assertions. The final script hash and runtime are recorded with the result; no parent or external checker cases are included.

No source correction is required by the local calculations reported here. In particular, the source already retains noncommuting normalizations, inverse-transpose derivatives, the all-order factorial base, square-summable pinching, and an all-pairs tail argument. The earned contribution is the explicit local relaxation and budget refinement at two retained interfaces, together with a precise inexact-update boundary. Classical tools are credited and the global geometric construction remains an imported claim.

References

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[6] M. Subbarao. *TR207: RMT robust calibration and inversion*. Confidential original local TeX, selected proof passages stated in Section 6. No public URL or external transmission.

[7] M. Subbarao. *Ram Calculus, Volume 21, Physical Metrics*, original `ch01d-physical-metrics.tex`, lines 1–210. Confidential local source.

[8] M. Subbarao. *TR-core-pat-16*, original inverse-sensitivity and rank body, lines 375–487. Confidential local source; not a surrogate for the separate TR207 and physical-metric proof readings.